

Personal, Social and Humanities Education Key Learning Area

**Economics Curriculum and Assessment Guide (Secondary 4-6)–
Supplementary Document
(with updates in 2015)
Effective from S4 in 2015/16**

Prepared by CDC-HKEAA Committee on Economics (Senior Secondary)

Introduction

This is a supplementary document to the Economics Curriculum. It is prepared by CDC-HKEAA Committee on Economics (Senior Secondary) in November 2015 for the purpose of enhancing teachers' understanding of the requirements of each topic. Teachers and students are suggested to use it alongside with the Economics Curriculum and Assessment Guide (Secondary 4-6) (with updates in November 2015) jointly prepared by Curriculum Development Council and the Hong Kong Examinations and Assessment Authority.

General notes to the document

- (1) The first three columns are extracted from Chapter 2 of the Economics Curriculum and Assessment Guide (Secondary 4-6) (updated in November 2015 (i.e. sections 2.2.1 and 2.2.2) where the update of the terms in Topic F made under the New Academic Structure (NAS) Medium-term Review is incorporated.
- (2) The part titled “elaboration of the curriculum contents” indicates the knowledge, concepts and skills that students are expected to acquire.
- (3) P.48 of this document outlines the arrangements of the Public Examination of Economics from 2016 onwards.

Compulsory Part

	Topic	Key points	Elaboration of the curriculum contents
A	Basic Economic Concepts	Economics as a social science	- Students are expected to understand and illustrate with examples that economics is a study of human behaviours, i.e. how individuals and societies decide on the use of scarce resources by referring to the postulate of constrained maximization. - Students are NOT expected to grasp the concepts on methodology (scientific method) in the study of economics.
		Scarcity, choice and opportunity cost (i) The source of economic problems: scarcity - Unlimited wants and limited resources - Free and economic goods (ii) Choice and opportunity cost - Economic decisions	- Students are expected to understand the - meaning of scarcity, - relationship between scarcity and choice, - relationship between scarcity, competition and discrimination, and - differences between scarcity and shortage. - Students are expected to - understand the meaning of good, the definitions of free good and economic good, and - be able to give examples of free good and economic good.

	Topic	Key points	Elaboration of the curriculum contents
		involving choices among alternatives • Concept of cost in economics • Interest as the cost of earlier availability of resources	• Students are expected to understand the definition of opportunity cost and explain (i) the relationship between choice and cost, (ii) why only the highest-valued option forgone is counted as cost, and (iii) why resource spent in the past such as past expenditure, which is irrecoverable, is not a cost though the term 'sunk cost' is NOT required. • Students are expected to (i) identify and calculate the cost involved in choosing an option, i.e. identify and distinguish explicit cost and implicit cost, and (ii) explain whether and how cost changes under different circumstances. • Students are expected to understand that decision making on the use of resources may involve an inter-temporal choice. From this perspective, they should be able to explain (i) why interest can be regarded as an opportunity cost (i.e. the cost of earlier availability of resources), and (ii) how interest rate, present consumption and saving are related. • Students are NOT expected to calculate present value. • Students are expected to understand the relationship between interest rate, investment and money demand. However, it could be discussed under the topics of macroeconomics.

	Topic	Key points	Elaboration of the curriculum contents
		The three basic economic problems (i) What to produce? How to produce? For whom to produce? (ii) How society tackles the basic economic problems • By society's customs and traditions • By government decisions • By the market mechanism (N.B. Illustrations by examples only. All theories on types of economic systems NOT required) (iii) Private property rights and its importance in a market economy	• Students are expected to (i) understand the meaning of each basic economic question, and (ii) be able to illustrate them with examples. • Students are expected to give examples of each method in tackling these economic problems. • Students are expected to (i) understand the meaning of private property rights - exclusive right to use, - exclusive right to receive income, - right to transfer, and (ii) explain why private property rights are important in a market economy: - clear delineation of private property rights is a prerequisite for the use of price mechanism

	Topic	Key points	Elaboration of the curriculum contents
			by which resources could be allocated to the highest-valued users.
		Specialization and exchange • Exchange as a condition for specialization Circular flow of economic activities (i) Consumption of households and production of firms (ii) The relationship among production, income and expenditure	• Students are expected to understand that exchange is a condition for specialization. [Note: The discussion of circular flow of economic activities can be scheduled in different ways. For instance, it can be treated as an introduction to Topic F “Measurement of Economic Performance”.] • Students are expected to (i) understand and sketch a well-labeled circular flow diagram for a closed economy, and (ii) understand the relationship among production, income and expenditure.
		Positive and normative statements • Distinction between positive statements and normative statements	• Students are expected to (i) understand the characteristics of positive statements and normative statements, for instance, - whether value judgment is involved, - whether they are refutable by facts, and hence disagreement to them can be settled by appealing to facts, (ii) give examples of normative and positive statements, and (iii) express their normative views to economic issues, conduct positive analysis on these views, and make informed judgment on the arguments/policies on the economic issues.

	Topic	Key points	Elaboration of the curriculum contents
B	Firms and Production	Ownership of firms (N.B. Firm as a unit that makes decisions regarding the employment of factors of production and the production of goods and services) (i) Forms of ownership • Public ownership • Private ownership: sole proprietorship, partnership and limited company (N.B. Classification of partnership NOT required)	• Students are expected to understand the meaning of firms. • Students are expected to (i) understand the meaning of public ownership, and (ii) give examples of organizations or institutions in Hong Kong operating under public ownership. • Students are expected to (i) classify a private firm into sole proprietorship, partnership, private limited company or public limited company, and (ii) understand the key features including legal status, liability, number of owners, etc. of different forms of private ownership. • Students are expected to apply their knowledge in this topic by comparing the features, and hence the advantages and disadvantages, of different types of ownership when they are given a concrete problem / situation. • Students are NOT expected to grasp the documentation involved in setting up business units.

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		(ii) Limited and unlimited liability (iii) Shares and bonds as sources of capital (N.B. Classification of shares and bonds NOT required)	- Students are expected to (i) understand the meaning of limited liability and unlimited liability, and (ii) identify the types of ownership that the owners could enjoy limited liability. - Students are expected to understand the (i) key differences between shares and bonds, (ii) advantages and disadvantages of issuing bonds and shares in raising capital, and (iii) advantages and disadvantages of buying bonds and shares from small investors' point of view.
		Types/stages of production Primary, secondary and tertiary production and their inter-relationship	- Students are expected to (i) understand the meaning of primary, secondary and tertiary production, (ii) give examples of different types of production, (iii) classify a given industry into primary, secondary or tertiary production, (iv) describe their inter-relationship, and (v) recognise the contributions of each type of production to the Hong Kong economy.
		Types of goods and services produced (i) Producer and consumer goods	- Students are expected to (i) understand that production involves production of goods and / or provision of services, (ii) understand the meaning of producer and consumer goods, and (iii) give examples of producer and consumer goods.

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		(ii) Private and public goods (N.B. Modelling regarding public goods NOT required)	- Students are expected to - understand the definition of private good, - understand that public good is defined as a good which is non-rival and non-excludable in consumption, and - give examples and distinguish whether a given good is private good or public good. [Note: Impure public good is NOT required.]
		Division of labour (i) Types: simple, complex and regional (ii) Advantages and disadvantages (iii) Limitations	- Students are expected to - understand the meaning of simple, complex and regional division of labour, - give examples of simple, complex and regional division of labour, - identify the advantages and disadvantages of division of labour, and - recognise the limitations of division of labour.
		Factors of production (i) Human resources Labour: supply, productivity, mobility and different methods of wage payments	- Students are expected to understand the - meaning of labour supply which is measured in terms of man-hour per time period, - factors that affect labour supply such as wages, population, population structure, etc., - meaning of and be able to calculate average labour productivity, - factors that affect labour productivity such as education, training, capital endowments, etc., - meaning of geographical and occupational mobility, - factors that affect geographical and occupational labour mobility, and

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		• Entrepreneurship: risk-bearing and decision-making (ii) Natural resources • Land: supply (iii) Man-made resources • Capital: accumulation and depreciation (iv) The features of (i) to (iii) in Hong Kong	(vii) advantages and disadvantages of different methods of wage payments. • Students are expected to understand special roles of an entrepreneur: risk-bearing and decision-making. • Students are expected to recognise that supply of natural resources could not be changed by human efforts. • Students are expected to understand the (i) meaning of capital accumulation, which involves giving up present consumption for future consumption, (ii) relationship between capital accumulation and interest rate, and (iii) meaning of depreciation. • Students are expected to describe the features of human resources, natural resources and man-made resources in Hong Kong.
		Production and costs in the short run and long run (i) Definition of short run and long run • In terms of fixed and	• Students are expected to understand the meaning and give examples of fixed and variable factors of production.

	Topic	Key points	Elaboration of the curriculum contents
		variable factors of production (ii) Law of diminishing marginal returns • Illustration by total product, average product and marginal product schedules only (iii) Cost of production • Fixed and variable costs • Total, marginal and average cost of production (N.B. General relationship between total, marginal and average cost curves NOT required. Relationship between short run and long run	• Students are expected to (i) state the law of diminishing marginal returns, (ii) convert marginal product to average product and total product, average product to marginal product and total product, total product to marginal product and average product NUMERICALLY, and (iii) illustrate the law of diminishing returns by total product, average product and marginal product schedules. • Students are NOT expected to grasp the relationship between the law of diminishing returns and U-shaped marginal cost curve. • Students are expected to understand the meaning and give examples of fixed and variable costs. • Students are expected to convert marginal cost to average cost and total cost, average cost to marginal cost and total cost, and total cost to marginal cost and average cost NUMERICALLY.

	Topic	Key points	Elaboration of the curriculum contents
		curves NOT required) (iv) Economies and diseconomies of scale • Internal economies and diseconomies of scale • External economies and diseconomies of scale (N.B. Economies and diseconomies of scale illustrated by average cost. Further classification of internal and external economies and diseconomies of scale NOT required) (v) Expansion and integration of firms • Types: vertical, horizontal, lateral and conglomerate • Motives	• Students are expected to (i) understand the meaning of internal / external economies and diseconomies of scale, (ii) illustrate internal / external economies and diseconomies of scale by average cost, and (iii) give possible reasons leading to internal / external economies and diseconomies of scale. • Students are expected to (i) understand the meaning and give examples of each type of integration, and (ii) give some possible motives behind each type of integration. • Students are NOT expected to grasp the methods of integration such as takeover and merger.
		The objectives of firms: (i) Profit maximization with	• Students are expected to recognise that the discussion is in the context of a price taking firm given that profit is POSITIVE.

	Topic	Key points	Elaboration of the curriculum contents
		given prices and marginal cost schedule • Meaning of profit as the difference between total revenue and total cost • Profit maximizing choice of output for individual firms with given prices and marginal cost schedule • The marginal cost schedule as the supply schedule of individual firms (N.B. Long run supply NOT required) (ii) Other objectives: market share, provision of non-profit making services, corporate social responsibility, etc	• Students are expected to explain why the marginal cost schedule of an individual firm is its supply schedule. • Students are NOT expected to grasp (i) the shut down point, breakeven point and long run supply of a price taking firm, (ii) the graphical relationship between MC, AC, AVC and supply curve, and (iii) that a monopolist does not have a supply curve. • Students are expected to grasp the concepts by using SCHEDULE ONLY. However, to cater for learners' diversity, teachers may use curves to explain the concepts if they find students could master the curves. • Students are expected to illustrate these objectives with examples.
C	Market and	Law of Demand	• Students are expected to

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	Price		(i) state the law of demand, and (ii) explain phenomena by applying the law of demand (using relative price or full price), such as how an imposition of lump-sum fee alters the relative price between two goods. [Note: Students are NOT expected to use the term “Alchian’s Generalisation”.]
		Individual demand (i) Factors affecting individual demand (ii) Complements and substitutes, superior and inferior goods (N.B. Giffen goods NOT required) (iii) Individual demand schedule and importance of the <i>ceteris paribus</i> assumption (iv) Difference between change in quantity demanded and change in demand	• Students are expected to (i) understand the meaning of complements, substitutes, superior and inferior goods, and (ii) explain how price of related goods, income, price expectation, and weather, etc. affect individual demand. • Students are expected to distinguish between change in demand (caused by changes in exogenous variables) and change in quantity demanded (caused by changes in an endogenous variable, i.e. the price of the good).

	Topic	Key points	Elaboration of the curriculum contents
		Market demand (i) Horizontal summation of individual demand curves (ii) Factors affecting market demand	- Students are expected to (i) understand that market demand could be obtained by horizontal summation of individual demand for a private good, (ii) understand the meaning of horizontal summation, and (iii) be able to add up the individual demand curves / schedules horizontally to obtain the market demand curve / schedule. - Students are expected to explain how price of related goods, income, price expectation, weather, and number of consumers in the market, etc. affect market demand.
		Individual supply (i) Factors affecting individual supply (ii) Individual supply schedule and importance of the <i>ceteris paribus</i> assumption (iii) Difference between change	- Students are expected to (i) state the law of supply, (ii) explain how price of related goods, cost of production, technology, price expectation and weather, etc. affect individual supply, and (iii) understand the meaning of joint supply and competitive supply. - Students are expected to distinguish between change in supply (caused by changes in exogenous variables) and change in quantity supplied (caused by changes in an endogenous variable, i.e. the price of the good).

	Topic	Key points	Elaboration of the curriculum contents
		in quantity supplied and change in supply	
		Market supply (i) Horizontal summation of individual supply curves (ii) Factors affecting market supply	- Students are expected to be able to add up the individual supply curves / schedules horizontally to obtain the market supply curve / schedule. - Students are expected to explain how price of related goods, cost of production, technology, price expectation, weather, and number of sellers in the market, etc. affect market supply.
		Interaction between demand, supply and price (i) Definition of equilibrium: no tendency to change (ii) Equilibrium price and quantity (iii) Effects of change in demand and/or change in supply on equilibrium price and quantity	- Students are expected to identify the (i) equilibrium price and quantity graphically, and (ii) equilibrium price and quantity, given the demand and supply schedules. - Students are expected to explain the effects of change in demand and/or change in supply on equilibrium price and quantity, with the aid of diagram(s).
		Consumer and producer surplus (i) Marginal benefit to	- Students are expected to (i) state the definition of consumer surplus, and

	Topic	Key points	Elaboration of the curriculum contents
		consumers, willingness to pay, consumer surplus, demand curve and their relationship (ii) Marginal cost of firms, minimum supply-price, producer surplus, supply curve and their relationship (iii) Illustrate consumer surplus and producer surplus in a demand-supply diagram (N.B. Concepts of utility, marginal rate of substitution, and indifference curves NOT required)	(ii) show consumer surplus on a (supply-)demand diagram. - Students are expected to treat “willingness to pay” as the same as “marginal benefit” to consumers. - Students are NOT expected to grasp the methods for extracting consumer surplus. [Note: The term “willingness to pay” is used in the curriculum. The term “marginal use value” is NOT recommended.] - Students are expected to (i) state the definition of producer surplus, and (ii) show producer surplus on a (demand-)supply diagram.

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		Functions of prices (i) Rationing function: existing supplies are distributed to users with highest value (ii) Allocative function • Demand is derived from marginal benefit, and supply is derived from marginal cost; the interaction between demand and supply then determines price and resources allocation • Changes in relative prices and resource deployment (N.B. Graphical analysis NOT required)	• Students are expected to understand the function of prices in the case of exchange without production, i.e. prices would direct existing endowments to highest-valued users. • Students are expected to understand that changes in relative prices would lead to changes in quantity of the goods produced, which implies changes in resource deployment.
		Price elasticity of demand (i) Arc elasticity	• Students are expected to calculate the (arc) price elasticity of demand, i.e. the average price and quantity method.

	Topic	Key points	Elaboration of the curriculum contents
		(N.B. Point elasticity, cross elasticity and income elasticity NOT required) (ii) Relationship between price elasticity and total revenue (iii) Factors affecting price elasticity of demand	- Students are expected to explain the relationship among price change, price elasticity and total revenue. - Students are expected to explain how the price ranges, availability of substitutes, degree of necessity, time, durability, proportion of income spent on the good and number of uses, etc. affect the price elasticity of demand. - Students are NOT expected to grasp the second law of demand, but they are expected to understand that price elasticity of demand is affected by the time for adjustment.
		Price elasticity of supply (i) Arc elasticity (N.B. Point elasticity and cross elasticity NOT required) (ii) Factors affecting price elasticity of supply	- Students are expected to calculate the (arc) price elasticity of supply, i.e. the average price and quantity method. - Students are expected to explain how factor mobility, flexibility of production and time, etc. affect the price elasticity of supply.
		Market intervention	Students are expected to

	Topic	Key points	Elaboration of the curriculum contents
		(i) Price intervention: price ceiling and price floor (ii) Quantity intervention: quota • Illustration of quota by a kinked supply curve (iii) Unit tax and unit subsidy • Determination of the share of the tax burden/subsidy between producers and consumers (N.B. Graphical illustration of price ceiling, price floor, quota, unit tax and unit subsidy and their impact on price and	(i) understand the meaning of (effective) price ceiling and (effective) price floor, and (ii) analyse the effects of the imposition of (effective) price ceiling / floor on (money) price, quantity transacted, consumer surplus and producer surplus, etc., with the aid of diagram(s). • Students are expected to (i) understand the meaning of (effective) quota, and (ii) analyse the effects of the imposition of (effective) quota on price, quantity transacted, consumer surplus, producer surplus and product quality, etc., with the aid of diagram(s). [Note: - The good is assumed to be homogenous, i.e. single homogenous product. - The analysis of effects of quota on imported goods of a small open economy is covered in Topic J.] • Students are expected to (i) determine the price, quantity transacted and the share of tax burden / subsidy between producers and consumers under the imposition of a unit tax / provision of a unit subsidy, (ii) analyse the effects of the imposition of unit tax / provision of unit subsidy on price, quantity transacted, consumer surplus and producer surplus, etc., with the aid of diagram(s), (iii) show the share of tax burden / subsidy between producers and consumers on a supply-demand diagram, and (iv) understand how price elasticities of demand and supply affect the distribution of tax burden / subsidy.

	Topic	Key points	Elaboration of the curriculum contents
		quantity)	
D	Competition and Market Structure	Perfect competition and imperfect competition (monopolistic competition, oligopoly and monopoly) (i) Definition of market (ii) General features • Number of sellers • Number of buyers • Nature of product • Ease of entry • Availability of information • Price taker/price searcher (iii) Sources of monopoly power (N.B. The four different forms of market structure are theoretical constructs. Actual examples may only be approximations of the above constructs. General analysis with marginal revenue and	• Students are expected to (i) understand that market is any arrangement that transactions take place, (ii) give examples of monopolistic competition, oligopoly and monopoly, and (iii) explain the sources of monopoly power such as natural monopoly, high set-up cost, legal entrance restrictions and public ownership, etc.

	Topic	Key points	Elaboration of the curriculum contents
		marginal cost curves NOT required)	
E	Efficiency, Equity and the Role of Government	Efficiency (i) Conditions for efficiency: Maximization of total social surplus; marginal benefit equals marginal cost (ii) Deviations from efficiency: • Price ceiling, price floor, tax, subsidy and quota • Deadweight loss (iii) Divergence between private and social costs (benefits): market versus government solutions, illustrated by examples ONLY	• Students are NOT expected to (i) use the term “Pareto condition”, and (ii) grasp the concept that it is always efficient when all constraints are considered. [Note: Inefficiency is possible in a partial analysis.] • Students are expected to (i) analyse the effects of price ceiling, price floor, tax, subsidy and quota on consumer surplus, producer surplus and efficiency, with the aid of supply-demand diagram(s), and (ii) understand the meaning of deadweight loss and indicate the deadweight loss on supply-demand diagram(s). [Note: The above sources that lead to deviations from efficiency are NOT exhaustive.] • Students are expected to (i) give examples that involve a divergence between private and social costs (benefits), (ii) explain how divergence emerges when they are given an example, (iii) explain the problem(s) of divergence between private and social costs (benefits) by referring to the existence of over-production / consumption (under-production / consumption),

	Topic	Key points	Elaboration of the curriculum contents
		(N.B. Graphical analysis with illustration of consumer surplus and producer surplus in a demand-supply diagram only. The term “Pareto condition” NOT required)	(iv) explain how government policies (e.g. taxes and subsidies) could be used to tackle the problems of divergence, and (v) explain how the problems of divergence could be tackled through market mechanism, i.e. negotiation and compensation (or payment) between the parties causing the external effects and the parties affected by these effects. - Students are NOT expected to grasp the Coase Theorem. [Note: Students are ONLY expected to discuss the problem(s) of divergence between private and social costs (benefits) by referring to examples. However, to cater for learners’ diversity, teachers may use diagrams to explain the concepts if they find students could master the curves.]
		Equity (i) Efficiency and equity in a market economy	- Students are expected to understand that (i) the discussion of equity in Topic E is limited to income inequality, (ii) resource allocation under perfectly competitive market situation is efficient, (iii) resource allocation through market mechanism is inefficient under certain circumstances (e.g. the existence of divergence between private and social costs (benefits) may lead to inefficiency), and (iv) even if resource allocation is efficient, there is an issue of income inequality. - Students are NOT expected to explain (i) the efficiency issue of public good even though they are expected to know the definition of public good, and (ii) in the Compulsory Part the efficiency implication of monopoly / price searching. [Note: A philosophical exploration of the general concept of equity is NOT required.]

	Topic	Key points	Elaboration of the curriculum contents
		(ii) Measuring income inequality: income distribution, Lorenz curve and Gini coefficient (N.B. Construction of the Lorenz curve and Gini coefficient NOT required) (iii) Sources of income inequality: human capital (e.g. skill differentials), discrimination and unequal ownership of capital, etc	- Students are expected to (i) understand the relationship between the size of Gini-coefficient and the shape of Lorenz curve, (ii) interpret the information delivered from the measuring tools: income distribution, Lorenz curve and Gini-coefficient, and (iii) analyse the issue of income inequality in Hong Kong. - Students are NOT expected to grasp detailed analysis of the labour market such as deriving labour demand from marginal revenue product.
		Policy concerns (i) Equalizing income or equalizing opportunities	- Students are expected to understand (i) that equity is a broad concept and so the discussion of policy impact on equity in this part is LIMITED to two commonly considered ethical principles: equalizing income and equalizing opportunities, and (ii) the normative choice of these principles would affect policy choices.

	Topic	Key points	Elaboration of the curriculum contents
		(ii) Disincentive effects of taxes and transfers (iii) Trade-off between equity and efficiency	- Students are expected to explain why some policy instruments that aim at equalizing income (e.g. taxes and transfers) may have disincentive effects. - Students are expected to discuss, with reasoning provided, the effects on income inequality (both in terms of income and opportunities) and efficiency when a policy is given. - Students are NOT expected to discuss THEORETICALLY the possibility of trade-off between equity and efficiency.
F	Measurement of Economic Performance	National income (i) National income as a general term for aggregates like Gross Domestic Product (GDP) and Gross National Income (GNI) (ii) Gross Domestic Product (GDP) The three approaches to measure GDP: production approach (value-added approach), income approach, expenditure approach	- Students are expected to grasp the (i) meaning of GDP, (ii) concept of resident producing units (RPU), (iii) items included / excluded in calculating GDP, (iv) use of the circular flow model to explain why GDP can be measured by 3 different approaches, (v) expenditure approach: $Y \equiv C + I + G + NX$,

	Topic	Key points	Elaboration of the curriculum contents
		(N.B. Components of GDP compiled under the income approach NOT required) • Nominal and real GDP • GDP at factor cost • per capita GDP; growth rate of GDP	- classification of the components, - GDP by expenditure components in HK, and (vi) production / value-added approach: - sum of value-added of all production activities by RPU's, - value-added = value of output – intermediate consumption. - GDP by economic activities in HK. [Note: Students are NOT expected to grasp the identity $S-I = NX$.] • Students are expected to understand the (i) meaning of nominal GDP (GDP at current market prices), (ii) problems of using nominal GDP, and (iii) meaning of real GDP. [Note: The Census and Statistics Department has adopted the method of chain volume measures to compile the data of real GDP. Teachers are advised to introduce the concept of chain volume measures of GDP to students to enhance their skill of interpreting local authentic data. Students are NOT expected to grasp the respective compilation method.] • Students are expected to understand the (i) meaning of GDP at factor cost, and (ii) relationship between GDP at factor cost and GDP at market prices. • Students are expected to understand the (i) meaning of per capita (nominal / real) GDP, and

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		(N.B. Other measures related to GDP NOT required) (iii) GNI as GDP plus net income from abroad (N.B. Other measures related to GNI NOT required) (iv) Uses and limitations of national income statistics as an indicator of economic welfare and for international comparison (N.B. Human Development Index NOT required)	(ii) growth rate of nominal GDP, real GDP, per capita nominal GDP and per capita real GDP, etc. - Students are NOT expected to grasp Net Domestic Product (NDP). - Students are expected to illustrate net factor income from abroad with examples. [Note: The Census and Statistics Department has adopted the term “net external primary income flows” to represent “net income from abroad”. Teachers are advised to introduce the term “net external primary income flows” to students to enhance their understanding of the local authentic data.] - Students are NOT expected to grasp GNI at constant market prices, GNI at factor cost and NNI, etc.

	Topic	Key points	Elaboration of the curriculum contents
		General price level as measured by Consumer Price Index and implicit price deflator of GDP (N.B. Construction of CPI and implicit price deflator of GDP NOT required)	- Students are expected to - calculate the rate of change in the general price level (inflation rate) by using CPI and GDP deflator, and - compare CPI and GDP deflator in terms of coverage and weighting, etc.
		Unemployment and underemployment rates as measured in terms of the percentage of unemployed and underemployed persons in the labour force	- Students are expected to understand the meaning of the - unemployment rate: $(\text{number of unemployed} / \text{labour force}) \times 100\%$, - unemployed in economics (that is, those who want a job but failed to find one), - underemployment rate: $(\text{number of underemployed} / \text{labour force}) \times 100\%$, and - underemployed in economics (that is, those who involuntarily work less than specified working hours). - Students are NOT expected to grasp the technical definitions of unemployed (underemployed) in compiling unemployment (underemployment) rate statistics.
		Recent trends of national income, general price level and unemployment in Hong Kong	Students are expected to interpret numerical and graphical data of the Hong Kong economy.

	Topic	Key points	Elaboration of the curriculum contents
G	National Income Determination and Price Level	Aggregate demand (AD) (i) Reasons for a downward sloping AD curve (ii) Determinants of aggregate demand: • Private consumption expenditure, which in turn depends on disposable income, the desire to save, wealth (value of assets), interest rate, etc • Investment expenditure, which in turn depends on business prospect, interest rate, etc • Government expenditure • Net export, which in turn depends on the economic conditions of trading	• Students are expected to grasp the three factors, i.e. wealth effect, interest rate effect and exchange rate effect, that lead to the downward sloping feature of an AD curve. • Students are expected to grasp (i) the factors that cause a shift in an AD curve, (ii) that private consumption, investment and import expenditures are positively dependent on income, and (iii) that one unit increase in disposable income will induce a less-than-one-unit increase in consumption expenditure. [Note: Students are NOT expected to grasp the mathematical function of consumption, investment and net-export.]

	Topic	Key points	Elaboration of the curriculum contents
		partners, exchange rate, etc (N.B. Derivation of the AD curve, magnitude of the shift in the AD curve and factors affecting the slope of the AD curve NOT required)	
		Aggregate supply (AS) (i) Reasons for an upward sloping short run AS curve (ii) Reasons for a vertical long run AS curve (iii) Factors affecting short run	• Students are expected to grasp “imperfect adjustment of input and output prices” as the explanation for an upward-sloping SRAS curve. • Students are expected to understand that in the long run (i) prices are flexible in a way that factor and product prices are fully adjustable, and (ii) the economy has capacity constraint. • Students are expected to understand the

	Topic	Key points	Elaboration of the curriculum contents
		and long run AS (N.B. Explanation by the Phillips curve and magnitude of the shift of the AS curve NOT required)	(i) factors affecting SRAS: - cost shocks, - government policies (changes in taxes, subsidies and regulations, etc.), and - other factors, and (ii) factors affecting LRAS: - factor endowments (labour, capital, natural resources) and technological changes, etc.
		The determination of level of output and price (i) Determination of the equilibrium level of output and price level in the AS-AD model (N.B. Quantity Theory of Money NOT required) (ii) Changes in the equilibrium level of output and price level caused by change(s) in the AD and/or AS	• Students are expected to grasp the comparative statics analysis (i.e. comparing the initial and final equilibria) in the (i) short run, and (ii) long run. However, they are NOT expected to describe the mechanism / adjustment process. • Students are expected to grasp the mechanism of moving from a short-run equilibrium to a long-run equilibrium.

	Topic	Key points	Elaboration of the curriculum contents
		(iii) Relationship between employment and output level (N.B. Interest rate is treated as exogenously determined)	- Students are expected to understand that a given event (change in any exogenous variable / factor / policy) MAY affect the AD and/or SRAS in the short run, and even affect the LRAS in the long run. In this case, students are expected to (i) specify whether they conduct a short run or long run analysis, and (ii) explain how the event leads to changes in AD, SRAS and/or LRAS by specifying - the component(s) of AD affected, and/or - factor(s) affected leading to a shift of SRAS / LRAS curve.
H	Money and Banking	Money (i) Definition of money (ii) Nature and functions of money	
		Banks: functions and services (i) Commercial banks and	- Students are expected to grasp the (i) functions (in particular, channeling savings to investment) and services provided by

	Topic	Key points	Elaboration of the curriculum contents
		central bank (ii) Licensed banks, restricted licence banks and deposit-taking companies in Hong Kong (iii) How central banking functions are performed in Hong Kong	commercial banks, and (ii) functions of a central bank. Students are expected to grasp the features of these three types of financial institutes.
		Money supply (i) Definitions of money supply in Hong Kong (ii) Credit creation/ contraction and the banking multiplier	- Students are expected to grasp the definitions of M1, M2 and M3 in Hong Kong. - Students are expected to - understand and calculate - required and actual reserve ratio, - monetary base, - maximum and actual banking multiplier, and - maximum and actual change in money supply, deposits and loans, etc.,

	Topic	Key points	Elaboration of the curriculum contents
			(ii) grasp the assumptions made in calculating the maximum changes in money supply, deposits and loans etc., and (iii) describe the process of credit creation / contraction. [Note: (i) Students are expected to understand that a change in monetary base will affect money supply, deposits and loans, etc.(ii) Students are NOT expected to manipulate cash-deposit ratio in the calculation of credit creation / contraction.]
		Money demand (i) Meaning of transactions demand for money and asset demand for money (ii) Money demand as a function of nominal interest rate and income	- Students are expected to understand the meaning of and factors affecting - transaction demand for money, and - asset demand for money. - Students are expected to (i) explain why demand for money depends positively on income, (ii) understand that cost of holding money is the nominal interest rate, (iii) explain why demand for money depends negatively on nominal interest rate, and (iv) identify the return of holding money and other benefits of holding money.
		Determination of interest rate in the money market Interaction of money supply and money demand	[Note: Students are NOT expected to grasp alternative models of explaining the determination of interest rate such as loanable fund theory.]

	Topic	Key points	Elaboration of the curriculum contents
		Hong Kong as a financial centre (i) Factors contributing to its development as a financial centre (ii) Effects on the Hong Kong economy	
I	Macroeconomic Problems and Policies	Business cycles: a description of the short run fluctuations in the real GDP around the long run trend (N.B. Theories of business cycles NOT required)	Students are expected to grasp the features of the 4 phases of a business cycle.
		Inflation and deflation (i) Definitions of inflation and deflation (ii) Relationship between nominal and real interest rates	Students are expected to understand the equation: $\text{nominal interest rate} = \text{real interest rate} + \text{expected inflation rate}.$

	Topic	Key points	Elaboration of the curriculum contents
		(iii) Redistributive effects (iv) Inflation and Quantity Theory of Money (N.B. Velocity of circulation of money assumed to be constant)	- Students are expected to explain and illustrate with examples, why debtors will gain (lose) and creditors will lose (gain) under unanticipated inflation (deflation) by comparing - the purchasing power of money in different time periods, and - expected real interest rate and realised real interest rate. - Students are expected to explain how the wealth of holders of real and monetary assets will be affected under inflation (deflation). - Students are expected to understand the meaning of the Equation of Exchange, i.e. $MV \equiv PY$ and hence, the meaning of velocity of circulation (V). - Students are expected to explain the changes in price level and nominal income and calculate inflation rate by using the quantity theory of money assuming (i) only V to be constant, or (ii) both V and Y to be constant.
		Unemployment (i) Meaning of unemployment (ii) Meaning of	- Students are expected to understand the meaning of (i) unemployed in economics, i.e., those who want a job but failed to find one (see Topic F), (ii) deflationary (output) gap, i.e. equilibrium output is less than full employment when the economy has an excess supply of labour, and (iii) inflationary (output) gap, i.e. equilibrium output is greater than full employment when the economy has an excess demand for labour. - Students are expected to understand the meaning of underemployed in economics, i.e. those who

	Topic	Key points	Elaboration of the curriculum contents
		underemployment (iii) Cost of unemployment (N.B. Philips curve NOT required)	involuntarily work less than specified working hours (see Topic F). • Students are expected to understand the cost of unemployment to (i) the unemployed, and (ii) society.
		Fiscal policy (i) Meaning of fiscal policy • Definition of budget; surplus budget, deficit budget and balanced budget • Taxation - Principles	• Students are expected to understand (i) Adam Smith's four taxation principles: - Economy, - Equity, - Certainty, - Convenience, and (ii) the taxation principle in Hong Kong, i.e., source principle. [Note: The term used by the Inland Revenue Department of the HKSAR Government is “the territorial source principle”.]

	Topic	Key points	Elaboration of the curriculum contents
		- Classification of taxes - Direct and indirect taxes - Progressive, proportional and regressive taxes • Public expenditure: classification by function (N.B. With specific reference to Hong Kong) (ii) Effect of fiscal policy on the level of output and price	• Students are expected to understand the meaning of direct and indirect taxes and illustrate with examples. • Students are expected to understand the (i) meaning of progressive, proportional and regressive taxes, and (ii) effects of these types of taxes on income inequality (See Topic E). • Students are expected to grasp (i) the classification of public expenditure into Education, Social Welfare or Health etc. and, (ii) that the size of public sector can be measured by the portion of public expenditure in GDP. • Students are expected to (i) explain whether a fiscal policy / given budget is expansionary or contractionary, (ii) explain the effects of an expansionary fiscal policy (e.g. deficit budget and same increase in expenditure and tax in government budget) and a contractionary fiscal policy (e.g. surplus budget) on price level and output with the AD-AS model, (iii) grasp the effects of taxation on individuals and firms, and (iv) grasp the socio-economic implications of changes in public expenditure.
		Monetary policy (i) Meaning of monetary policy	• Students are expected to understand (i) the effects of changes in money supply and interest rate, (ii) how the execution of monetary tools affects money supply and/or interest rate, and

	Topic	Key points	Elaboration of the curriculum contents
		(ii) Effect of monetary policy on the level of output and price	(iii) the monetary policy in Hong Kong. - Students are expected to (i) explain whether a monetary policy is expansionary or contractionary, and (ii) explain the effects of an expansionary and a contractionary monetary policy on price level and output with the AD-AS model.
J	International Trade and Finance	Free trade and trade barriers (i) Absolute advantage, comparative advantage and gains from trade (N.B. Illustration by the production possibilities frontier NOT required) (ii) Using the pattern of trade in Hong Kong to illustrate the principle of comparative advantage (iii) Importance of trade to Hong	- Students are expected to (i) understand the meaning of absolute advantage and comparative advantage, (ii) explain which country has an absolute advantage in the production of a good / goods in a given case, (iii) explain which country has a comparative advantage in the production of a good in a given case, (iv) understand the condition(s) in which mutually beneficial trade is possible; (v) calculate the gains from trade, and (vi) state the principle of comparative advantage. - Students are expected to illustrate the importance of trade to Hong Kong with reference to given

	Topic	Key points	Elaboration of the curriculum contents
		Kong's economy (iv) Trade barriers • Types • Effects of tariff and quota on price and output for a small open economy • Trade barriers faced by Hong Kong (v) Hong Kong's attempts to overcome trade barriers • Trade promotion • Role of the HKSAR Government • Role of international economic institutions (e.g. World Trade Organisation)	data. • Students are expected to understand different types of trade barriers including tariff, quota, import surcharge, and embargo, etc. • Students are expected to analyse the effects of tariff and quota on price, domestic output, import and consumption quantity, and tariff revenue (if applicable) for a small open economy.
		Brief introduction to the balance of payments account • Current account - Main components of the	• Students are expected to understand (i) the meaning of credit and debit in the Balance of Payments Accounting; (ii) whether and how a given transaction affects the current account; (iii) the meaning of trade surplus and deficit, and

	Topic	Key points	Elaboration of the curriculum contents
		current account: goods, services, income and current transfers (N.B. Sub-classification of these components NOT required) • Capital and financial account (N.B. Sub-classification of this account NOT required)	(iv) the meaning of current account surplus and deficit. [Note: (i) Trade balance refers to visible trade balance in the curriculum and assessment, though the Census and Statistics Department has ceased to publish visible trade balance statistics. It publishes statistics of balance of trade in goods and balance of trade in services. (ii) The Census and Statistics Department has adopted the change of ownership principle in the compilation of current account. Teachers are advised to introduce this principle to students to enhance their understanding of the economic situation and trade pattern of Hong Kong.] • Students are expected to understand (i) whether and how a given transaction affects the capital and financial account, and (ii) the meaning and role of foreign and official reserve in the Balance of Payments Accounting. • Students are expected to understand (i) that Balance of Payments is always balanced in accounting sense, and (ii) the meaning of Balance of Payment surplus and deficit.
		Exchange rate (i) Meaning of exchange rate (N.B. Graphical analysis NOT required)	• Students are expected to understand the meaning of (i) exchange rate, and (ii) appreciation and depreciation, revaluation and devaluation.
		(ii) Effect of a change in the exchange rate on import	• Students are expected to grasp the effects of a change in exchange rate on (i) import and export prices in terms of domestic and foreign currencies,

	Topic	Key points	Elaboration of the curriculum contents
		price and export price	(ii) import and export volume, and (iii) values of import and export in terms of domestic and foreign currencies.
		(iii) Brief introduction to the linked exchange rate system in Hong Kong (N.B. Mechanism of maintaining the linked exchange rate NOT required)	- Students are expected to grasp the (i) brief history of the linked exchange rate system, and (ii) note-issuing mechanism (Currency Board System) in Hong Kong.

Elective Part 1

	Topic	Key points	Elaboration of the curriculum contents
	Monopoly Pricing	(i) Simple monopoly pricing • Determination of price and output • Efficiency implications (N.B. Graphical and numerical illustrations with given demand, marginal revenue and marginal cost curves) (ii) Price discrimination • Meaning of price discrimination • Types: First, second and third degree price discrimination • Conditions for different types of	• Students are expected to (i) understand and illustrate numerically the relationship between price and marginal revenue under simple monopoly pricing, (ii) show the relationship between demand curve and marginal revenue curve graphically, (iii) determine profit-maximizing output and price graphically and numerically, (iv) explain why simple monopoly pricing is inefficient, i.e. marginal benefit is greater than marginal cost, and illustrate with consumer surplus, producer surplus and deadweight loss, and (v) compare the output, price and efficiency implications between monopoly under uniform pricing and perfectly competitive market. • Students are expected to give examples of different types of price discrimination. [Note: Students are NOT expected to grasp efficiency implications of different types of price discrimination.]

	Topic	Key points	Elaboration of the curriculum contents
		price discrimination (N.B. Price and output determination NOT required)	
	Anti-competitive Behaviours and Competition Policy	(i) Major forms of anti-competitive practices • Horizontal agreements among competitors: agreements to restrict prices and output • Vertical agreements between buyers and sellers • Mergers: horizontal mergers, vertical mergers and potential competition mergers (ii) The impact of anti-competitive	• Students are expected to understand anti-competitive behaviours under horizontal agreements among competitors including price fixing, collusive bidding / bid rigging, market division / market allocation, customer allocation and sales and production quotas, etc. • Students are expected to understand anti-competitive behaviours under vertical agreements between buyers and sellers including resale price maintenance (RPM), tie-in sales and exclusive dealing, etc. • Students are expected to explain some possible impacts of different types of anti-competitive practices, such as higher price, lower output, lack of choices to consumers, and reduction of the number of

	Topic	Key points	Elaboration of the curriculum contents
		practices (N.B. Graphical analysis NOT required) (iii) The Competition Ordinance in Hong Kong	competitors, etc. - Students are expected to grasp (i) the objectives of the Competition Ordinance, (ii) the first conduct rule covering agreements, concerted practices and decisions that prevent, restrict or distort competition, (iii) the second conduct rule covering abuse of market power; and (iv) exclusions and exemptions.

Elective Part 2

	Topic	Key points	Elaboration of the curriculum contents
	Extension of Trade Theory	(i) Illustration of comparative costs and gains from trade with the aid of production possibilities frontier (N.B. The use of indifference curve NOT required) (ii) Comparative advantage and its relation to globalization	- Students are expected to (i) understand the slope of production possibilities frontier as marginal cost of producing good X, (ii) illustrate comparative advantage with the aid of production possibilities frontiers and/or terms of trade, (iii) determine production point, and (iv) show the gains from trade. [Note: Students are NOT expected to determine consumption point by using indifference curve.] - Students are expected to understand (i) the meaning of globalization (focus on the economic aspects), and (ii) comparative advantage and its relation to globalization, for example, the Mainland as ‘the world factory’.
	Economic Growth and Development	(i) Measurement of economic growth and development • Changes in real GDP • Changes in per capita real GDP • Changes in Human Development Index	- Students are expected to (i) understand the three dimensions of Human Development Index, and

(ii) Factors affecting growth of an economy

- Inputs: physical capital, human capital, natural resources, technological change
- Policies: saving and investment, foreign direct investment, trade, education, population, property rights, research and development

(N.B. The analytical framework of aggregate production function and the theories and models of economic growth NOT required)

(ii) interpret the Human Development Index.

- Students are expected to describe the effects of these factors on the growth of an economy.
- Students are expected to interpret graphical and numerical data related to economic growth of an economy.

	(iii) The desirability and costs of economic growth • Trade-off between current and future consumption • Growth, living standard and income distribution • Resources exhaustion, pollution and sustainable development (iv) International/regional comparison	• Students are expected to discuss the effects of economic growth with reference to data. • Students are expected to interpret graphical and numerical data related to economic growth of different economies.
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Public Examination of Economics

The following tables show the arrangements of the Public Examination to be implemented from 2016 onwards.

2016-2018 Examinations

Papers	Components	Weight	Duration
Paper 1 (Compulsory Part)	Multiple-choice questions	30%	1 hour
Paper 2 (Compulsory and Elective Parts)	Conventional paper Part 1: Compulsory Part (compulsory) Part 2: Elective Part (select one out of two questions)	70%	2 hrs 15 mins

From 2019 HKDSE Examination onwards, the examination time of Paper 2 will be extended from 2 hours 15 minutes to 2 hours 30 minutes.

Papers	Components	Weight	Duration
Paper 1 (Compulsory Part)	Multiple-choice questions	30%	1 hour
Paper 2 (Compulsory and Elective Parts)	Conventional paper Part 1: Compulsory Part (compulsory) (Includes ONE data-response question) Part 2: Elective Part (select one out of two questions)	70%	2 hrs 30 mins

[Note: Although School-based Assessment is not implemented in Economics, teachers should continue organising activities to facilitate students' learning of Economics such as integrating and applying contents of various topics to enquire into daily-life and economic phenomena.]